FLATHEAD QUADRUPED -- WIRING DIAGRAM

+5V5_SERVO
+5V0_LOGIC
+3V3_ESP
+3V3_P1
I2C_SDA
I2C_SCL
UART_ESP_TX
UART_ESP_RX
I2S_BCLK
I2S_LRC
I2S_SD_MIC
I2S_SD_AMP
BATT_SENSE
GND

2S LiPo 7.4V 2200mAh

GND
VBAT_RAW
V+
BT1

ESP32 DevKit

VBAT_SENSE (GPIO34)
RX2 (GPIO16)
TX2 (GPIO17)
SCL (GPIO22)
SDA (GPIO21)
GND
3V3_OUT
5V
U3

MPU6050

SCL
SDA
GND
VCC 3V3
U4

Raspberry Pi Zero 2 W

SD_AMP_OUT (gpio40)
SD_MIC_IN (gpio38)
LRC (gpio35)
BCLK (gpio12)
CSI
RX (gpio15)
TX (gpio14)
GND (pin9)
3V3 (pin1)
5V (pin2)
U6

INMP441

SD
WS
SCK
L/R (GND)
GND
VDD 3V3
U8

1.8–3.3V ONLY -- 5V destroys it.

F1

VBAT_FUSED
VBAT_RAW
OUT
IN

20A Blade Fuse

SW1

VBAT_SW
VBAT_FUSED
OUT
IN

KCD1 Rocker Switch

XL4016 Buck Converter

GND
+5V5_SERVO
GND
VBAT_SW
VOUT−
VOUT+ 5.5V
VIN−
VIN+
U1

C1

4700uF 16V

UBEC

GND
+5V0_LOGIC
GND
VBAT_SW
VOUT−
VOUT+ 5.0V
VIN−
VIN+
U2

R2 / R1

GND
4k7
R2
BATT_SENSE
10k
R1
VBAT_SW

8.4V -> 2.7V at GPIO34

PCA9685 #1 (addr 0x40)

CH15
CH14
CH13
CH12
CH11
CH10
CH9
CH8
CH7
CH6
CH5
CH4
CH3
CH2
CH1
CH0
V+ (NC)
SCL
SDA
GND
VCC 3V3
U5

V+ = NC (12A trace rating) -- servos powered from +5V5_SERVO.

Servo Bank (legs + arm 1)

CH15
CH14
CH13
CH12
CH11
CH10
CH9
CH8
CH7
CH6
CH5
CH4
CH3
CH2
CH1
CH0
GND
V+ 5.5V
J1

OV5647 Camera Module

CSI
U7

MAX98357A

SPK−
SPK+
DIN
LRC
BCLK
GND
VIN 5V
U9

LS1

3W 4–8ohm Speaker

PCA9685 #2 (addr 0x41)

CH15
CH14
CH13
CH12
CH11
CH10
CH9 — LED2_B
CH8 — LED2_G
CH7 — LED2_R
CH6 — LED1_B
CH5 — LED1_G
CH4 — LED1_R
CH3
CH2
CH1
CH0
V+ (NC)
SCL — I2C_SCL
SDA — I2C_SDA
GND — GND
VCC 3V3 — +3V3_ESP
U10

Same I2C bus as PCA9685 #1. Bridge the A0 pad so this board answers at 0x41 (#1 stays at 0x40). V+ = NC, servos fed from +5V5_SERVO.

Arm 2 Servos

CH3 — gripper
CH2 — elbow
CH1 — pitch
CH0 — yaw
GND
+5V5_SERVO
V+ 5.5V
J2

RGB Eye LED 1 (common anode)

+5V0_LOGIC
LED1_R — 220 — R
LED1_G — 150 — G
LED1_B — 150 — B
A (+5V)
LED1

RGB Eye LED 2 (common anode)

+5V0_LOGIC
LED2_R — 220 — R
LED2_G — 150 — G
LED2_B — 150 — B
A (+5V)
LED2

Eye LEDs: anode to +5V0_LOGIC, each cathode through its resistor to a PCA9685 #2 channel. Red 220 ohm, green/blue 150 ohm (about 13 mA each). Run PCA9685 #2 in open-drain mode (MODE2 OUTDRV = 0) so the 3.3V-powered outputs can sink the 5V LED current.

Rev G: 2nd PCA9685 for arm 2, RGB eye LEDs, power chain on net labels
12-DOF legs + two 4-DOF arms -- ESP32 + Raspberry Pi Zero 2 W

Flathead Robotics

Sheet: /
File: Flathead_Wiring.kicad_sch

Title: FLATHEAD QUADRUPED — WIRING DIAGRAM

Size: A2	Date: 2026-10-21	Rev: G
KiCad E.D.A. 10.0.5		Id: 1/1